SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Purpose and Scope

TrackWIL is a web-based system proposed to improve the management of Work-Integrated Learning (WIL) documents and student progress. The system replaces the manual, paper-reliant process of completing and signing daily WIL documents, which often leads to lost records and processing delays. The scope of the platform includes centralizing the workflow so that students can digitally submit daily work records and evidence, workplace mentors can efficiently review and verify submissions, and academic coordinators and lecturers can seamlessly monitor overall student progress.

2. Users and Stakeholders

- **Students (Primary):** Need a simple, reliable platform to create daily WIL records, upload supporting documents, and check the verification status of their submissions without relying on email or paper.
- **Workplace Mentors (Primary):** Require an organized interface to view records submitted by assigned students, verify completed work, approve records, and request necessary corrections with feedback.
- **Coordinators (Primary):** Need an overview of student documentation to monitor submissions, identify missing records, and ensure WIL requirements are fulfilled.
- **Lecturers (Primary):** Need to track academic progression to identify students who are falling behind and provide necessary support.
- **Secondary Stakeholders:** Include the Training Institution (needs efficient record management), the Development Team (needs clear requirements and tools to build the system), and the Project Supervisor (oversees the project's academic alignment).

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3. Functional Requirements

- **FR-01:** The system must allow authorized users to log in using valid credentials. (Priority: High) | *Acceptance Criterion: User is granted access to the appropriate role-based dashboard upon entering correct credentials.*
- **FR-02:** The system must provide different access permissions for students, mentors, coordinators, and lecturers. (Priority: High) | *Acceptance Criterion: Users can only access features and data permitted by their designated roles.*
- **FR-03:** Students must be able to create daily WIL records describing the work completed. (Priority: High) | *Acceptance Criterion: The system provides an accessible form for students to input daily tasks.*
- **FR-04:** Students must be able to submit completed daily records for review. (Priority: High) | *Acceptance Criterion: Record successfully transitions to a "pending" state for mentor review upon submission.*
- **FR-05:** Students must be able to upload supporting documents or evidence related to their daily records. (Priority: High) | *Acceptance Criterion: The system accepts and securely attaches file uploads to the respective daily record.*
- **FR-06:** Workplace mentors must be able to view records submitted by their assigned students. (Priority: High) | *Acceptance Criterion: Mentors can access a populated list of pending records from their specific mentees.*
- **FR-07:** Workplace mentors must be able to approve records or request corrections with feedback. (Priority: High) | *Acceptance Criterion: The system allows mentors to attach feedback notes and update the record state accordingly.*
- **FR-08:** The system must display the status of each record as pending, approved, or requiring corrections. (Priority: High) | *Acceptance Criterion: Clear status indicators are visible on both the student and mentor dashboards.*
- **FR-09:** Students must be able to view their previous submissions and their status. (Priority: Medium) | *Acceptance Criterion: A history log is available for students to track past entries.*
- **FR-10:** Coordinators must be able to identify students with outstanding or missing records. (Priority: High) | *Acceptance Criterion: Coordinators have access to a reporting view highlighting unsubmitted mandatory records.*
- **FR-11:** Lecturers must be able to monitor student progress and identify students who are falling behind. (Priority: High) | *Acceptance Criterion: Lecturers can view aggregated metrics on student progression.*
- **FR-12:** The system must record the dates and times of submissions and verifications. (Priority: High) | *Acceptance Criterion: All record interactions generate accurate, automated timestamps.*
- **FR-13:** Authorized users must be able to view relevant student activity and progress information. (Priority: Medium) | *Acceptance Criterion: Dashboard information dynamically updates based on the viewing user's authorization level.*

- **FR-14:** Users must be able to log out of their accounts securely. (Priority: High) | *Acceptance Criterion: Active sessions are completely terminated upon logging out.*

4. Non-Functional Requirements

- **NFR-01 (Usability):** TrackWIL should feature a simple, understandable interface for seamless navigation, record submission, and status checking.
- **NFR-02 (Performance):** The system must load dashboard information and process document submissions within defined, acceptable response-time targets without unnecessary delays.
- **NFR-03 (Security):** The system must enforce role-based access control, secure document handling, and proper authentication mechanisms.
- **NFR-04 (Reliability):** TrackWIL must guarantee the reliable storage of submitted records and documents, utilizing backup and recovery measures to mitigate data loss risks.
- **NFR-05 (Maintainability):** The application will utilize an organized project structure and clear documentation to support developers in correcting errors, maintaining features, and updating the system over time.
- **NFR-06 (Compatibility):** The interface must adaptively support different screen sizes, including mobile phones, tablets, and computers, across commonly used modern web browsers.
- **NFR-07 (Data Integrity):** The system must maintain accurate data relationships, strictly linking submissions to correct placements and students, and restricting changes exclusively to authorized users.
- **NFR-08 (Privacy):** Access to WIL documents and personal information must be strictly limited to users who require the data to fulfill their specific responsibilities.

5. Use Cases / User Stories

- **US-01: Student Login:** As a student, I want to log in securely so that I can access my WIL records and dashboard.
- **US-02: Daily Record Submission:** As a student, I want to create and submit a daily WIL record so that my workplace activities are documented and stored in one place.
- **US-03: Upload Supporting Documents:** As a student, I want to upload supporting evidence so that my workplace activities can be reviewed without relying entirely on paper documents or email attachments.
- **US-05: Mentor Verification:** As a workplace mentor, I want to review student submissions and approve them or request corrections so that the records are properly checked.
- **US-06: Monitor Outstanding Records:** As a coordinator, I want to identify students with missing or outstanding records so that I can follow up and help ensure that the required documentation is completed.
- **US-07: Monitor Student Progress:** As a lecturer, I want to view relevant student progress information so that I can identify students who are falling behind and provide support.

6. Data Requirements

The system requires the management of fundamental data entities, including user accounts, WIL placements, daily activity records, document references, and chronological verification histories. Validation rules must ensure that each submission is properly linked to its corresponding student and workplace placement. Important system actions, such as submitting, approving, or requesting corrections on a record, must be documented to maintain an accurate document history. Data must be reliably retained; a relational database architecture (e.g., PostgreSQL) will be used to maintain these complex relationships, while uploaded evidence files will be held in a secure, access-restricted file storage service.

7. Interfaces

- **User Interface (UI) / Presentation Layer:** A web-based front-end that provides specific, role-based dashboards for students, mentors, coordinators, and lecturers. It will handle all direct interactions, such as record creation, file uploading, and status monitoring.
- **Application Interface / Business Logic Layer:** Sits between the UI and database to process logic, authenticate sessions, check role permissions, manage document verification rules, and prevent unauthorized actions (e.g., ensuring a student cannot approve their own submission as a mentor).
- **Database Interface / Data Layer:** Responsible for retrieving and saving records securely. When a request is validated by the application layer, this interface communicates with the relational database to commit the data and returns a success/failure response back to the user.

8. Constraints and Assumptions

- **Legal & Privacy Constraints:** The system must strictly adhere to South African data protection regulations, specifically the Protection of Personal Information Act (POPIA), ensuring that data is only collected for authorized WIL administration purposes and that user consent is handled appropriately.
- **Technical Constraints:** The development will follow an Agile System Development Life Cycle (SDLC) approach to iteratively test and implement features like login, submissions, and dashboard routing.
- **Operational Assumptions:** It is assumed that all stakeholders will have access to modern web browsers and an internet connection to utilize the cloud-based functionalities effectively. Mentors are assumed to manually verify the accuracy of work before approving, rather than relying on automated approvals.

9. Acceptance Criteria

The solution will be accepted when:

1. Students can successfully log in, fill out daily task records, upload corresponding evidence files, and transmit them for review.
2. Workplace mentors can successfully view assigned submissions, toggle statuses to approved, or return records with necessary correction feedback.
3. Coordinators and lecturers can successfully pull aggregated data to identify missing documentation and track academic pacing.
4. Role-based security is proven to restrict permissions effectively across the application layer, ensuring data privacy standards and POPIA compliance are met.
5. Iterative Agile testing confirms that all primary functional features load and operate without debilitating performance delays or data integrity loss.